

Unit IV Activity II

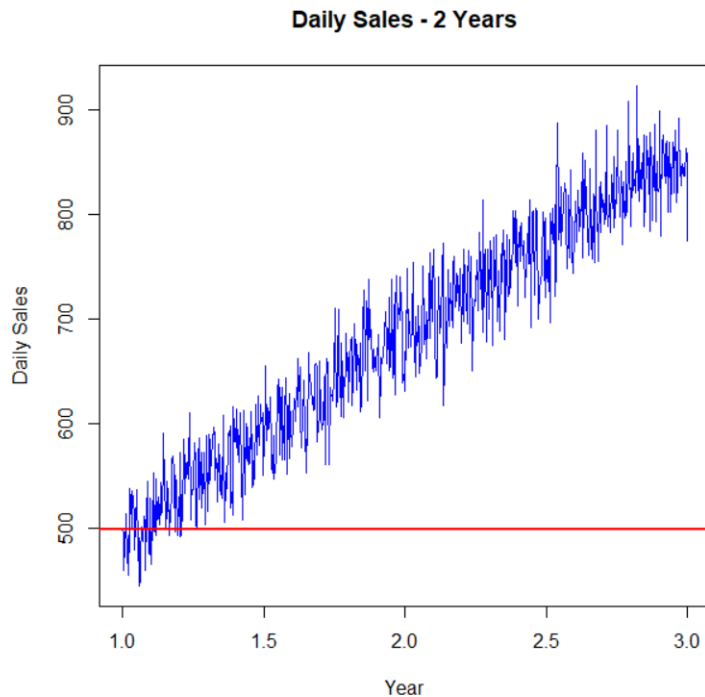
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COURSE CODE: DSA0604

1. Daily Sales Time Series

```
# Daily sales for 2 years
set.seed(10)
dates <- seq(as.Date("2024-01-01"),
             as.Date("2025-12-31"), by = "day")
sales <- 500 + 0.5 * (1:length(dates)) +
        rnorm(length(dates), 0, 30)
sales_ts <- ts(sales, frequency = 365)
plot(sales_ts,
     main = "Daily Sales - 2 Years",
     xlab = "Year",
     ylab = "Daily Sales",
     col = "blue")
abline(lm(sales ~ seq_along(sales)), col = "red", lwd = 2)
cat("The red line shows the long-term trend.\n")
```



2. Hourly Server Temperature

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
np.random.seed(10)
time = pd.date_range("2025-01-01", periods=100, freq="h")
temperature = 60 + np.random.normal(0, 3, 100)
temperature[30:40] += 12
temperature[70:80] += 15
df = pd.DataFrame({
    "Time": time,
    "Temperature": temperature
})
threshold = 70
plt.figure(figsize=(10, 5))
plt.plot(df["Time"], df["Temperature"], label="Temperature")
unsafe = df["Temperature"] > threshold
```

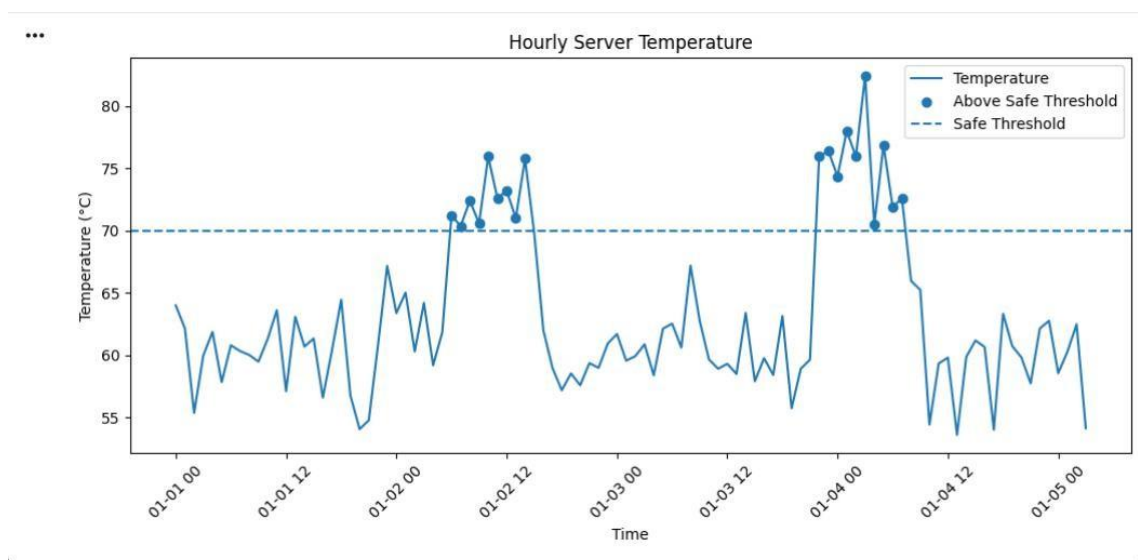
```

plt.scatter(df.loc[unsafe, "Time"],
            df.loc[unsafe, "Temperature"],
            label="Above Safe Threshold")

plt.axhline(threshold,
            linestyle="--",
            label="Safe Threshold")

plt.xlabel("Time")
plt.ylabel("Temperature (°C)")
plt.title("Hourly Server Temperature")
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



3. Three Tech Stocks

```
set.seed(20)

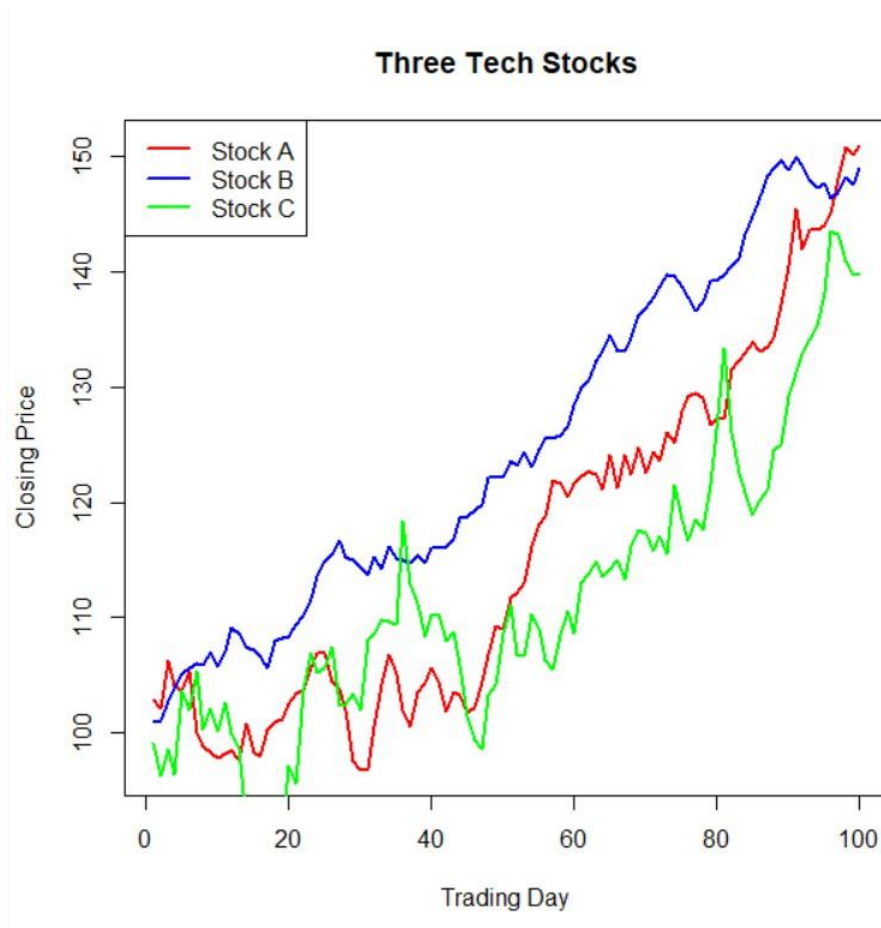
days <- 100

stock_A <- 100 + cumsum(rnorm(days, 0.5, 2))
stock_B <- 100 + cumsum(rnorm(days, 0.4, 1))
stock_C <- 100 + cumsum(rnorm(days, 0.3, 3))

plot(stock_A,
      type = "l",
      col = "red",
      lwd = 2,
      xlab = "Trading Day",
      ylab = "Closing Price",
      main = "Three Tech Stocks")

lines(stock_B, col = "blue", lwd = 2)
lines(stock_C, col = "green", lwd = 2)

legend("topleft",
      legend = c("Stock A", "Stock B", "Stock C"),
      col = c("red", "blue", "green"),
      lty = 1,
      lwd = 2)
```



4. Monthly Electricity Consumption

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

np.random.seed(30)

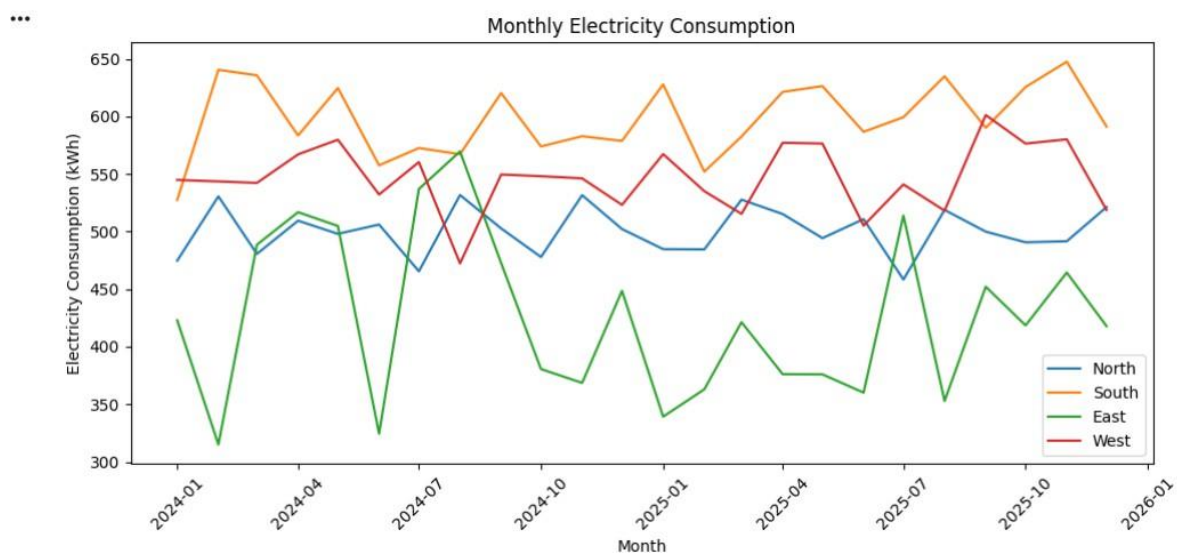
months = pd.date_range("2024-01-01", periods=24, freq="MS")
north = 500 + np.random.normal(0, 20, 24)
south = 600 + np.random.normal(0, 25, 24)
east = 450 + np.random.normal(0, 70, 24)
west = 550 + np.random.normal(0, 30, 24)

plt.figure(figsize=(10, 5))
plt.plot(months, north, label="North")
```

```

plt.plot(months, south, label="South")
plt.plot(months, east, label="East")
plt.plot(months, west, label="West")
plt.xlabel("Month")
plt.ylabel("Electricity Consumption (kWh)")
plt.title("Monthly Electricity Consumption")
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



5. Drug Dose–Response Curve

```

dose <- c(0, 10, 20, 30, 40, 50, 60, 80, 100)
response <- c(5, 15, 30, 48, 65, 78, 85, 89, 90)
plot(dose, response,
     pch = 19,
     col = "blue",
     xlab = "Drug Dose (mg)",
     ylab = "Patient Response (%)",
     main = "Drug Dose-Response Curve")

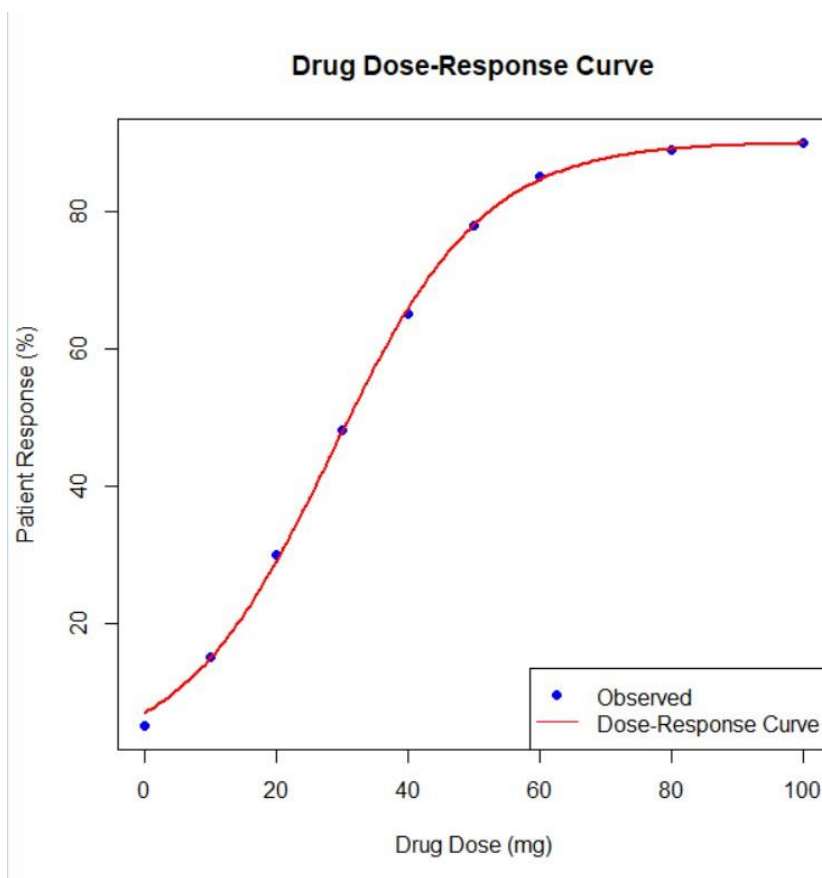
```

```

model <- nls(response ~ a/(1 + exp(-(dose-b)/c)),
             start = list(a = 90, b = 30, c = 10))
newdose <- seq(0, 100, length.out = 200)
predicted <- predict(model,
                    newdata = data.frame(dose = newdose))

lines(newdose, predicted,
      col = "red",
      lwd = 2)
legend("bottomright",
      legend = c("Observed", "Dose-Response Curve"),
      col = c("blue", "red"),
      pch = c(19, NA),
      lty = c(NA, 1))

```



6. Fertilizer Dose–Response

```
import numpy as np

import matplotlib.pyplot as plt

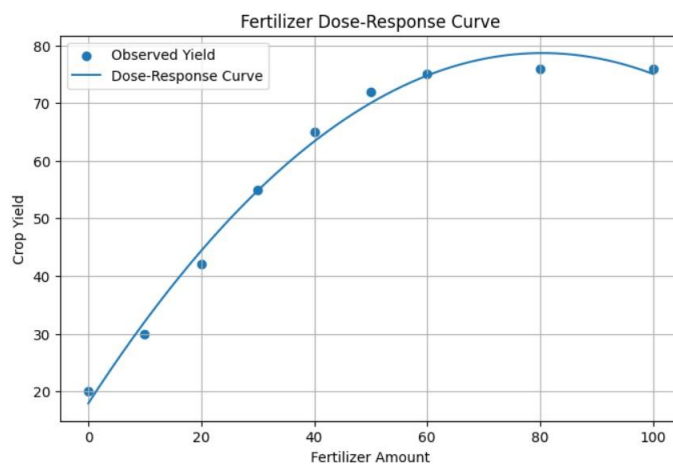
fertilizer = np.array([0, 10, 20, 30, 40, 50, 60, 80, 100])
yield_data = np.array([20, 30, 42, 55, 65, 72, 75, 76, 76])
coefficients = np.polyfit(fertilizer, yield_data, 2)
model = np.poly1d(coefficients)
x = np.linspace(0, 100, 200)
y = model(x)

plt.figure(figsize=(8, 5))
plt.scatter(fertilizer, yield_data,
            label="Observed Yield")

plt.plot(x, y,
         label="Dose-Response Curve")

plt.xlabel("Fertilizer Amount")
plt.ylabel("Crop Yield")
plt.title("Fertilizer Dose-Response Curve")
plt.legend()
plt.grid(True)
plt.show()
```

...



7. STL Decomposition of Monthly Ticket Sales

```
set.seed(50)

sales <- 1000 +
  5 * (1:36) +
  100 * sin(2 * pi * (1:36) / 12) +
  rnorm(36, 0, 30)

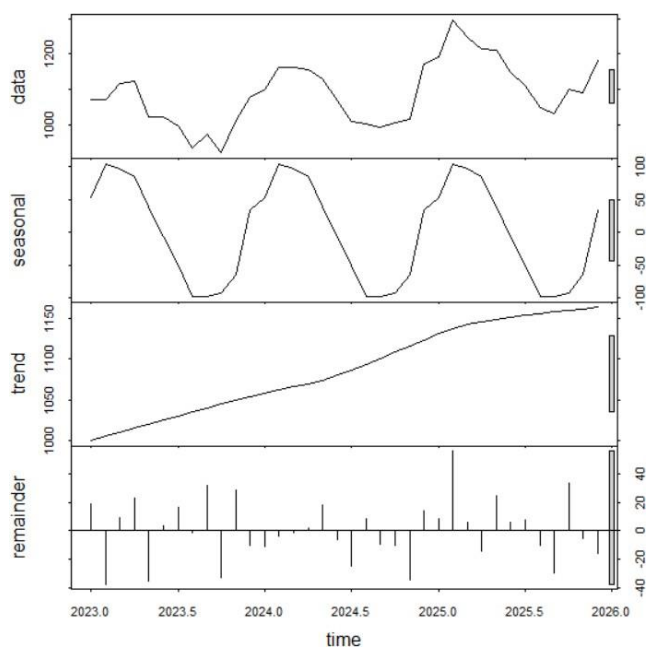
sales_ts <- ts(sales,
  start = c(2023, 1),
  frequency = 12)

decomp <- stl(sales_ts, s.window = "periodic")

plot(decomp)

trend <- decomp$time.series[, "trend"]
seasonal <- decomp$time.series[, "seasonal"]
remainder <- decomp$time.series[, "remainder"]

print(trend)
print(seasonal)
print(remainder)
```



8. STL Decomposition of Weekly Users

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import STL

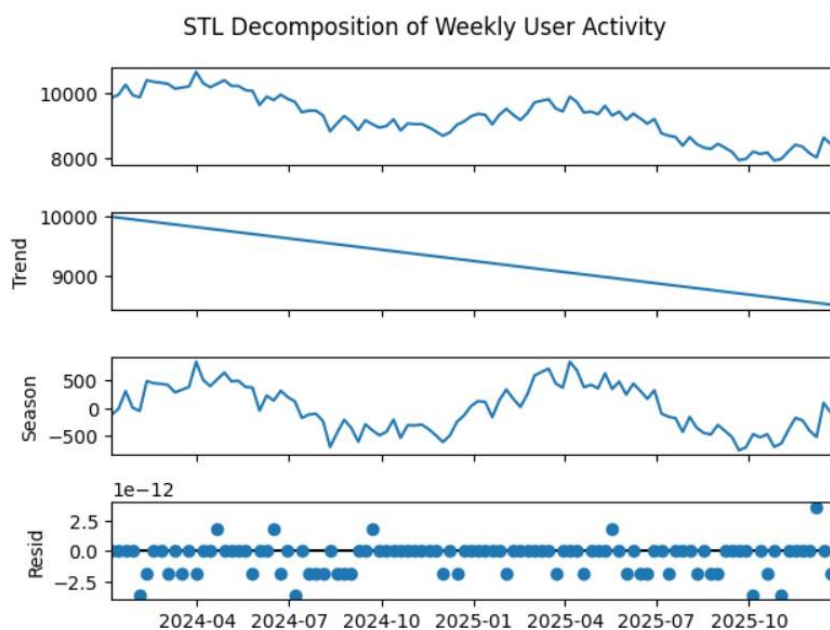
np.random.seed(60)

weeks = pd.date_range("2024-01-07",
                      periods=104,
                      freq="W")

users = (10000
        - 15 * np.arange(104)
        + 500 * np.sin(2 * np.pi * np.arange(104) / 52)
        + np.random.normal(0, 150, 104))

series = pd.Series(users, index=weeks)
result = STL(series, period=52).fit()
result.plot()

plt.suptitle("STL Decomposition of Weekly User Activity")
plt.tight_layout()
plt.show()
```



9. 7-Day and 30-Day Moving Average

```
set.seed(70)

price <- 100 + cumsum(rnorm(100, 0.2, 3))

MA7 <- filter(price,
               rep(1/7, 7),
               sides = 2)

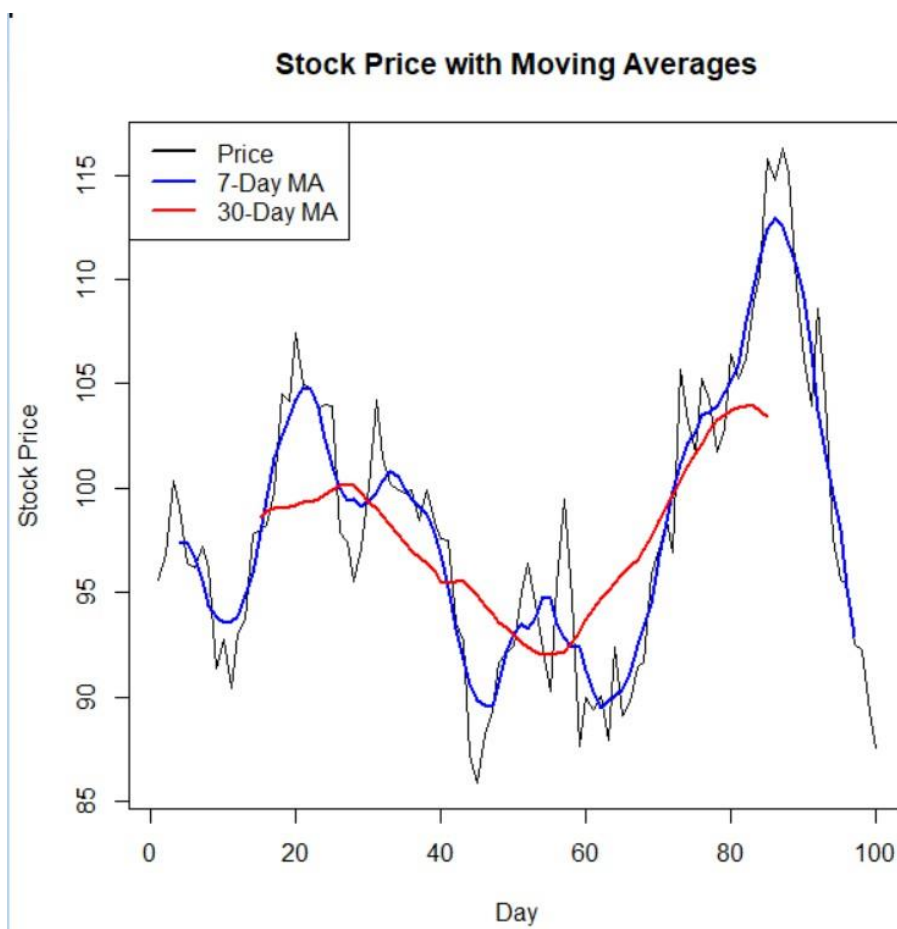
MA30 <- filter(price,
                rep(1/30, 30),
                sides = 2)

plot(price,
      type = "l",
      col = "black",
      xlab = "Day",
      ylab = "Stock Price",
      main = "Stock Price with Moving Averages")

lines(MA7,
      col = "blue",
      lwd = 2)

lines(MA30,
      col = "red",
      lwd = 2)

legend("topleft",
      legend = c("Price", "7-Day MA", "30-Day MA"),
      col = c("black", "blue", "red"),
      lty = 1,
      lwd = 2)
```



10. GDP Detrending

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
np.random.seed(80)
quarters = pd.date_range("2015-01-01",
                          periods=40,
                          freq="QS")
gdp = (20000
       + 150 * np.arange(40)
       + 500 * np.sin(2 * np.pi * np.arange(40) / 8)
       + np.random.normal(0, 100, 40))
df = pd.DataFrame({
    "Quarter": quarters,
```

```

"GDP": gdp
})

t = np.arange(len(df))

coefficients = np.polyfit(t, df["GDP"], 1)

trend = np.polyval(coefficients, t)

df["Detrended_GDP"] = df["GDP"] - trend

plt.figure(figsize=(10, 5))

plt.plot(df["Quarter"],

         df["Detrended_GDP"])

plt.axhline(0,

            linestyle="--")

plt.xlabel("Quarter")

plt.ylabel("Detrended GDP")

plt.title("Detrended Quarterly GDP")

plt.xticks(rotation=45)

plt.tight_layout()

plt.show()

```

